Skyler Ma

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Technical Skills

Languages/Frameworks: C, C++, Python, Go, TypeScript, HTML/CSS/JS, React.js, Django
Technologies: STM32, Arduino, KiCAD, Git, GitHub, Docker

Experience

Electrical Subsystem Member Sept 2025 – Present
Waterloo Rocketry *Waterloo, Ontario*

- Developed **embedded C** firmware for **STM32**-based avionics to read injector sensor data and transmit via **CAN**
- Designed PCB layout in **KiCAD** for remote arming system, enabling arming of recovery pyrotechnics using a magnetic switch, while integrating altimeter control, redundant power management, and **CAN bus** telemetry
- Soldered components onto PCBs, performed hardware debugging using a multimeter and oscilloscope

Full-Stack Software Development Co-op Jan 2026 – Present
Guelph Eye MD *Guelph, Ontario (Hybrid)*

- Developed a **Go** backend aggregating data from electronic medical record software via a **Python Java Access Bridge** API, in order to automate workflows and generate reports about clinic resources
- Built interfaces: a **React** web app, **Raspberry Pi Python** clients, and a **TypeScript** Stream Deck plugin
- Built a **CI/CD** pipeline using **GitHub Actions** and **Docker** to regularly deploy updates across **Windows and Ubuntu VMs on Hyper-V**, serving the web app over **internally-managed TLS** for everyday use by clinic staff

Full-Stack Software Development Intern Oct 2023 – June 2024
Prom Planner / Digitera Interactive *Ottawa, Ontario (Remote)*

- Developed **Laravel/PHP** web frontend and backend with role-based authentication for students, school admin, and businesses to efficiently manage prom registrations, payments, communications, and activities
- Conducted thorough app testing, documenting bugs and their causes, and recommending UI improvements

Projects

ESP32 Guitar Tuner (Prototype) | *C++, ESP32* Feb 2026 – Present

- Designed a single-transistor **amplifier** and **biasing** circuit to condition guitar signal for ADC input
- Wrote **embedded C++** firmware for **ESP32** to read audio signal and detect pitch in real time using a zero-crossing algorithm

Arduino Wearable Sleep Tracker | *C++, Arduino, Python* Oct 2025 – Nov 2025

- Developed **C++** firmware for a prototype of a wearable sleep tracker, reading accelerometer data from ADC and over **I2C**, detecting periodic motion using a cosine similarity algorithm, and transmitting serially over Bluetooth
- Developed **Python** software to receive, log, and plot data from a Bluetooth serial connection

Drama Attendance Manager | *PostgreSQL, Express.js, React, Next.js, Docker* Dec 2024 – June 2025

- Developed a full-stack web app to facilitate attendance tracking and management for students and teachers
- Used **Git/GitHub**, **Docker Compose**, and agile development for fast, organized, and adaptable collaboration

FPSwap — Managing Moving Patients | *Python, Django, HTML/CSS/JS, Vue.js* June 2022 – Jan 2024

- Developed a full-stack web app to help family physicians find new doctors for patients moving elsewhere in Ontario
- Created algorithm to efficiently find openings for incoming patients with doctors who had other patients leaving

Education

University of Waterloo Sept 2025 – Apr 2030
Candidate for Bachelor's of Applied Science in Computer Engineering, Honours, Co-op *Waterloo, Ontario*

- Cumulative GPA: 3.9/4.0